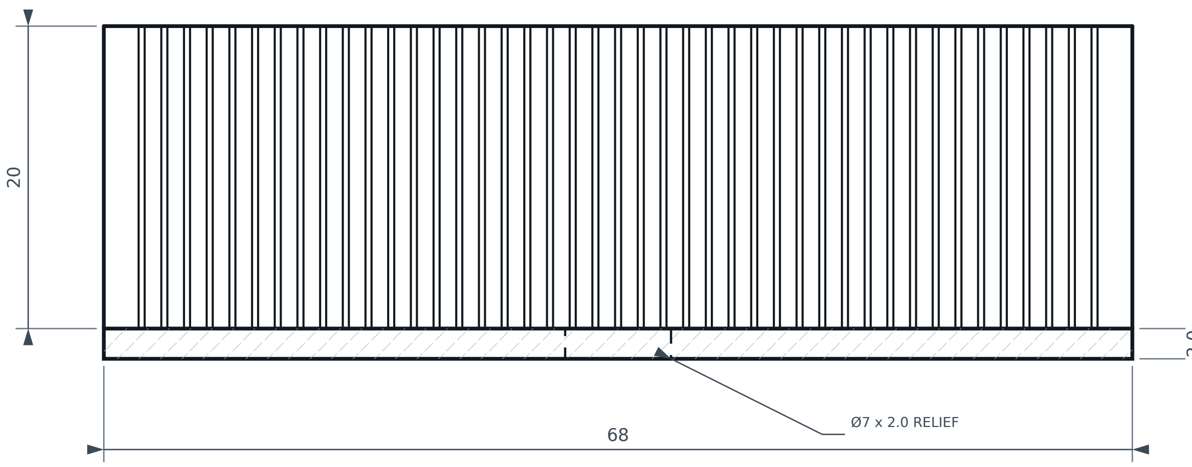
CONFIG A — VC-201

FORCED AIR >= 1 m/s

43 FINS x 0.40 THK @ 1.5 PITCH

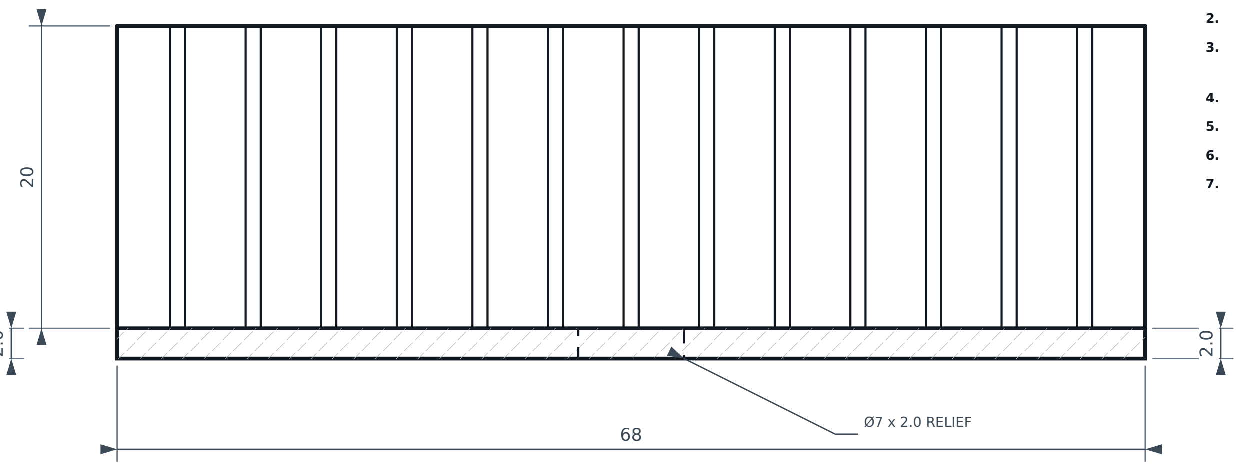


FRONT ELEVATION

CONFIG B — VC-202

NATURAL CONVECTION ONLY

13 FINS x 1.00 THK @ 5.0 PITCH



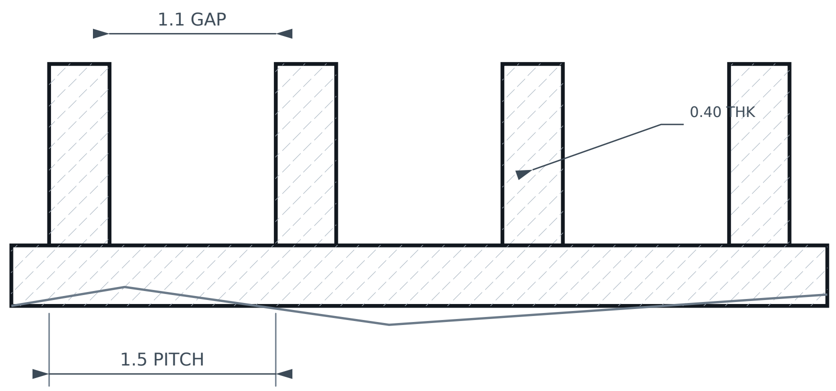
FRONT ELEVATION

NOTES

1. MATL: Al 1100-H14 or 6063-T5. Skived, folded or bonded fin.
2. Base flatness 0.03. Ra 0.8 max on the TIM face.
3. VC-202 SHALL be anodised, 5 um clear minimum. Radiation carries 28% of the passive load.
4. VC-201 finish is free - forced convection swamps radiation.
5. Fit ONE configuration only - alternates, not stackable.
6. Relief pocket clears the folded pinch-off tube, 0.5 min after pinch.
7. Fin tips deburred. No swarf.

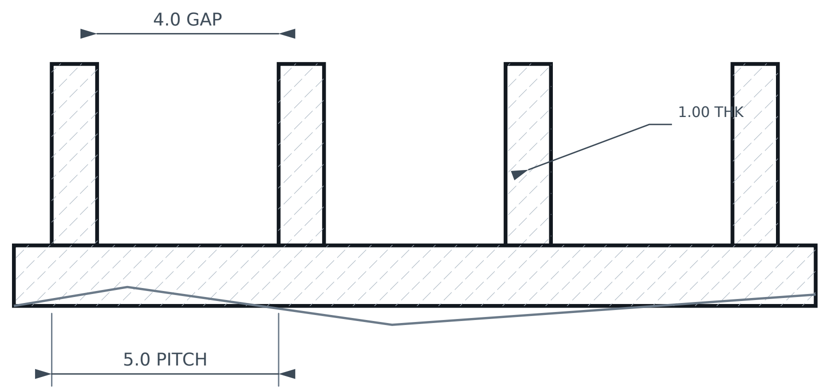
FIN DETAIL

SCALE 20:1



FIN DETAIL

SCALE 6:1



PASSIVE CAPACITY - VC-202, Tj 85 C MAX, NO FAN, FINS VERTICAL

AMBIENT	dT	FIN BASE	CONVECTION	RADIATION	TOTAL
25 C	60 K	82 C	11.9 W	4.6 W	**16.5 W**
30 C	55 K	82 C	10.5 W	4.3 W	14.8 W
35 C	50 K	82 C	9.1 W	4.0 W	13.1 W
40 C	45 K	83 C	7.7 W	3.7 W	11.4 W
45 C	40 K	83 C	6.4 W	3.4 W	9.8 W

WHY THE FINISH IS SPECIFIED - Tj 85 C, 25 C AMBIENT

FIN FINISH	eps	RAD	TOTAL	GAIN
bare mill finish	0.05	0.3 W	12.4 W	1.00x
chromate conversion	0.60	3.3 W	15.2 W	1.23x
clear anodise 5 um (SPECIFIED)	0.85	4.6 W	**16.5 W**	1.33x
black anodise	0.92	5.0 W	16.9 W	1.36x

Pitch optimum 4.8 mm (17.0 W); 5.0 mm used as a round tooling pitch.

Passive floor with channels fully blocked: 4.6 W from envelope radiation alone.

SELECTION TABLE - Tj 85 C ROWS AT 25 C AMBIENT

	PITCH	FINS	FINISH	Q @ Tj 100 C	Q @ Tj 85 C
VC-201 - forced air 1 m/s	**1.5**	**43**	any	**67 W**	**54 W**
VC-201 - forced air 2 m/s	**1.5**	**43**	any	**113 W**	**90 W**
VC-201 - forced air 4 m/s	**1.5**	**43**	any	**163 W**	**130 W**
VC-201 - STILL AIR (invalid)	1.5	43	anodised	9 W	7 W
VC-202 - passive, fins VERTICAL	**5.0**	**13**	**anodised**	**22 W**	**16.5 W**
VC-202 - passive, fins horizontal	5.0	13	anodised	13 W	9.5 W
VC-202 - passive, bare aluminium	5.0	13	bare	17 W	12.4 W

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE	SCALE	UNITS
FIN STACK — TWO CONFIGURATIONS Al 1100-H14 / SKIVED OR BONDED / 68 x 68 x 22	**2:1** THIRD ANGLE PROJECTION	**mm**

DRAWING No.	REV	SHEET
VC-201 / VC-202	**A**	**5 / 7**

DRAWN	CHECKED	MATERIAL	FINISH	DATE
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-09